
Chapter 6

Sample Run — Play Tennis Dataset

6.1 Dataset

The Play Tennis dataset is a classic benchmark for decision tree algorithms. It contains 14 samples described by four categorical attributes — Outlook, Temperature, Humidity, and Wind — with the target class PlayTennis (Yes/No). Table 6.1 shows the dataset structure:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Table 6.1: Play Tennis Dataset (14 Samples)

6.2 Output

```
Enter Excel dataset path: play_tennis.xlsx
```

```
=====
BUILDING ID3 DECISION TREE
=====
```

```
=====
ITERATION 1
=====
```

```
Current Node : ROOT
```

```
CURRENT DATASET USED IN THIS ITERATION:
```

```
-----
Outlook  Temp  Humidity  Wind  PlayTennis
  Sunny   Hot   High    Weak    No
  Sunny   Hot   High  Strong    No
Overcast  Hot   High    Weak    Yes
  Rain    Mild  High    Weak    Yes
  Rain    Cool  Normal  Weak    Yes
  Rain    Cool  Normal  Strong   No
Overcast  Cool  Normal  Strong   Yes
  Sunny   Mild  High    Weak    No
  Sunny   Cool  Normal  Weak    Yes
  Rain    Mild  Normal  Weak    Yes
  Sunny   Mild  Normal  Strong   Yes
Overcast  Mild  High    Strong   Yes
Overcast  Hot   Normal  Weak    Yes
  Rain    Mild  High    Strong   No
-----
```

Entropy of Dataset = 0.9403

Attribute: Outlook
Entropy(Sunny) = 0.971
Entropy(Overcast) = 0.0
Entropy(Rain) = 0.971
Gain(Outlook) = 0.2467

Attribute: Temp
Entropy(Hot) = 1.0
Entropy(Mild) = 0.9183
Entropy(Cool) = 0.8113
Gain(Temp) = 0.0292

Attribute: Humidity
Entropy(High) = 0.9852
Entropy(Normal) = 0.5917
Gain(Humidity) = 0.1518

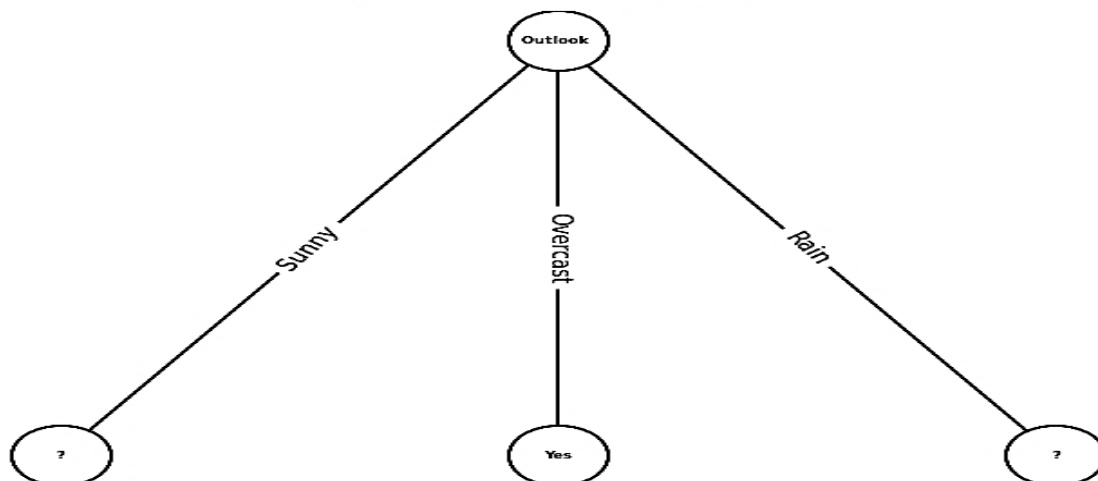
Attribute: Wind
Entropy(Weak) = 0.8113
Entropy(Strong) = 1.0
Gain(Wind) = 0.0481

Best Attribute Selected = Outlook

Tree Rules after iteration 1

IF Outlook = Sunny THEN ?
IF Outlook = Overcast THEN PlayTennis = Yes
IF Outlook = Rain THEN ?

Decision Tree After Iteration 1



=====

ITERATION 2

=====

Current Node : Outlook = Sunny

CURRENT DATASET USED IN THIS ITERATION:

Outlook	Temp	Humidity	Wind	PlayTennis
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes

Entropy of Dataset = 0.971

Attribute: Temp
Entropy(Hot) = 0.0
Entropy(Mild) = 1.0
Entropy(Cool) = 0.0
Gain(Temp) = 0.571

Attribute: Humidity
Entropy(High) = 0.0
Entropy(Normal) = 0.0
Gain(Humidity) = 0.971

Attribute: Wind
Entropy(Weak) = 0.9183
Entropy(Strong) = 1.0
Gain(Wind) = 0.02

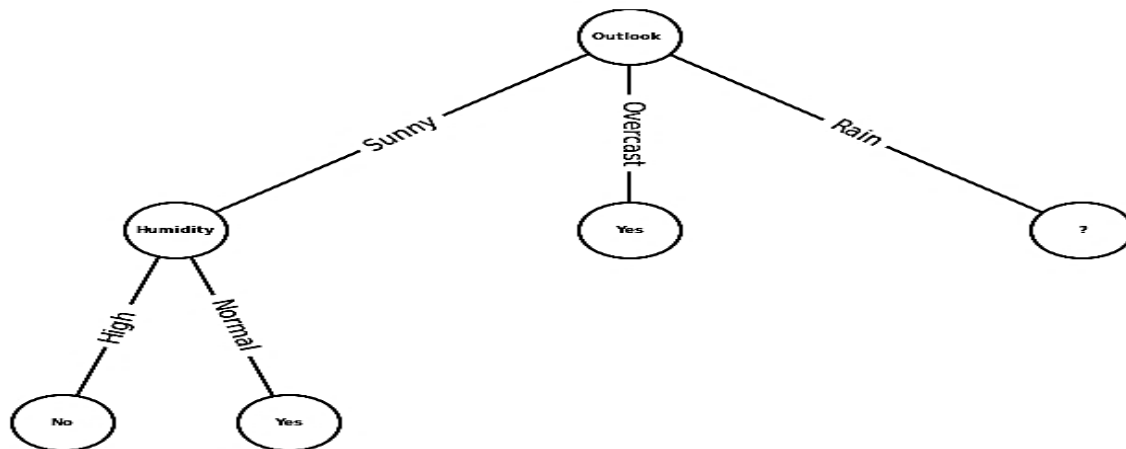
Best Attribute Selected = Humidity

Tree Rules after iteration 2

```

IF Outlook = Sunny THEN:
  IF Humidity = High THEN PlayTennis = No
  IF Humidity = Normal THEN PlayTennis = Yes
IF Outlook = Overcast THEN PlayTennis = Yes
IF Outlook = Rain THEN ?
  
```

Decision Tree After Iteration 2



ITERATION 3

Current Node : Outlook = Rain

CURRENT DATASET USED IN THIS ITERATION:

Outlook	Temp	Humidity	Wind	PlayTennis
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Rain	Mild	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Entropy of Dataset = 0.971

Attribute: Temp

Entropy(Mild) = 0.9183

Entropy(Cool) = 1.0

Gain(Temp) = 0.02

Attribute: Humidity

Entropy(High) = 1.0

Entropy(Normal) = 0.9183

Gain(Humidity) = 0.02

Attribute: Wind

Entropy(Weak) = 0.0

Entropy(Strong) = 0.0

Gain(Wind) = 0.971

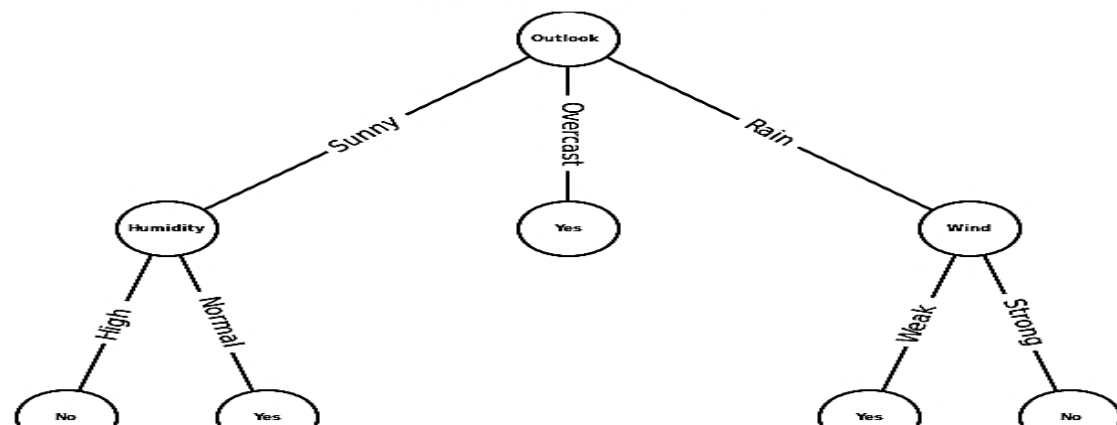
Best Attribute Selected = Wind

Tree Rules after iteration 3

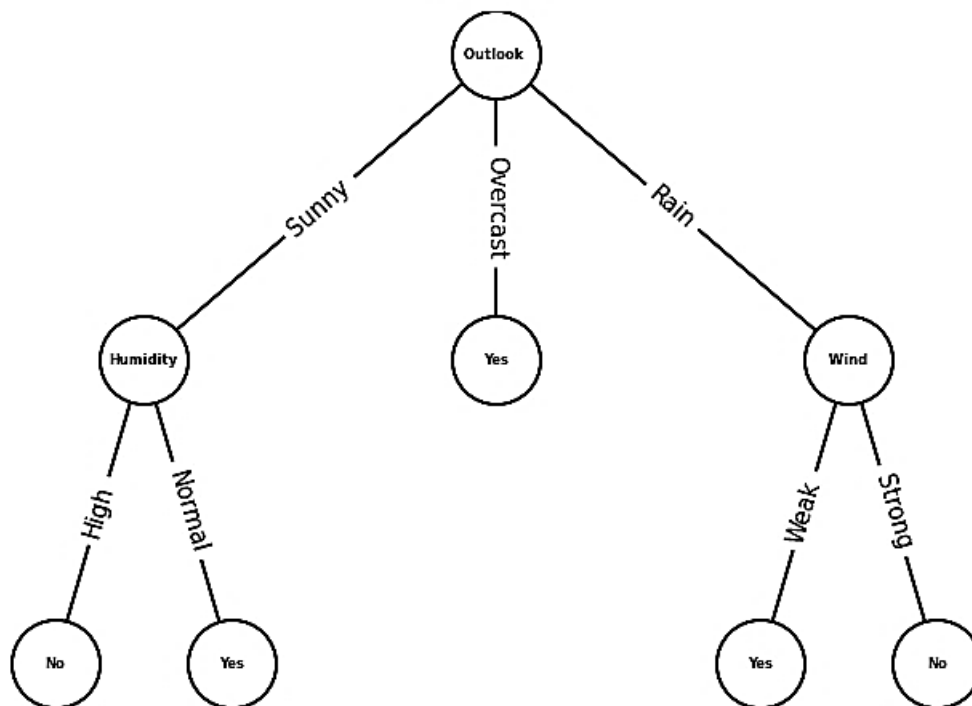
```

IF Outlook = Sunny THEN:
  IF Humidity = High THEN PlayTennis = No
  IF Humidity = Normal THEN PlayTennis = Yes
IF Outlook = Overcast THEN PlayTennis = Yes
IF Outlook = Rain THEN:
  IF Wind = Weak THEN PlayTennis = Yes
  IF Wind = Strong THEN PlayTennis = No
  
```

Decision Tree After Iteration 3



FINAL ID3 DECISION TREE



=====

ENTER VALUES TO CLASSIFY NEW SAMPLE

=====

Enter value for Outlook :sunny
Enter value for Temp :hot
Enter value for Humidity:high
Enter value for Wind :weak

=====

CLASSIFICATION RESULT

=====

New Sample: {'Outlook ': 'sunny', 'Temp ': 'hot', 'Humidity': 'high', ' Wind ': 'weak'}
Predicted Class: No

Chapter 7

Sample Run — Buy Computer Dataset

7.1 Dataset

The Buy Computer dataset contains 14 samples described by four attributes — Age, Income, Student, and Credit_Rating — with the target class BuysComputer (Yes/No). Table 7.1 presents the dataset:

RID	Age	Income	Student	Credit_Rating	BuysComputer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle_aged	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle_aged	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle_aged	Medium	No	Excellent	Yes
13	Middle_aged	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

Table 7.1: Buy Computer Dataset (14 Samples)

7.2 Output

```
Enter Excel dataset path: buys_computer.xlsx
```

```
=====
BUILDING ID3 DECISION TREE
=====
```

```
=====
ITERATION 1
=====
```

```
Current Node : ROOT
```

```
CURRENT DATASET USED IN THIS ITERATION:
```

```
-----
Age Income Student Credit_Rating Buys_Computer
<=30 High No Fair No
<=30 High No Excellent No
31...40 High No Fair Yes
>40 Medium No Fair Yes
>40 Low Yes Fair Yes
>40 Low Yes Excellent No
31...40 Low Yes Excellent Yes
<=30 Medium No Fair No
<=30 Low Yes Fair Yes
>40 Medium Yes Fair Yes
<=30 Medium Yes Excellent Yes
31...40 Medium No Excellent Yes
31...40 High Yes Fair Yes
>40 Medium No Excellent No
-----
```

```
Entropy of Dataset = 0.9403
```

```
Attribute: Age
```

```
Entropy(<=30) = 0.971
```

```
Entropy(31...40) = 0.0
```

```
Entropy(>40) = 0.971
```

```
Gain(Age) = 0.2467
```

Attribute: Income
 Entropy(High) = 1.0
 Entropy(Medium) = 0.9183
 Entropy(Low) = 0.8113
 Gain(Income) = 0.0292

Attribute: Student
 Entropy(No) = 0.9852
 Entropy(Yes) = 0.5917
 Gain(Student) = 0.1518

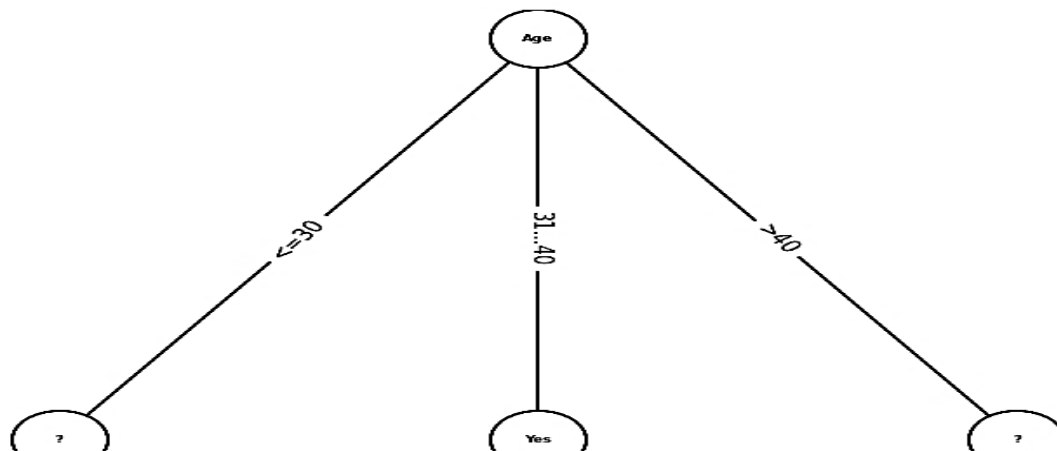
Attribute: Credit_Rating
 Entropy(Fair) = 0.8113
 Entropy(Excellent) = 1.0
 Gain(Credit_Rating) = 0.0481

 Best Attribute Selected = Age

Tree Rules after iteration 1

IF Age = <=30 THEN ?
 IF Age = 31...40 THEN Buys_Computer = Yes
 IF Age = >40 THEN ?

Decision Tree After Iteration 1



=====

ITERATION 2

=====

Current Node : Age = <=30

CURRENT DATASET USED IN THIS ITERATION:

Age	Income	Student	Credit_Rating	Buys_Computer
<=30	High	No	Fair	No
<=30	High	No	Excellent	No
<=30	Medium	No	Fair	No
<=30	Low	Yes	Fair	Yes
<=30	Medium	Yes	Excellent	Yes

Entropy of Dataset = 0.971

Attribute: Income
 Entropy(High) = 0.0
 Entropy(Medium) = 1.0
 Entropy(Low) = 0.0
 Gain(Income) = 0.571

Attribute: Student
 Entropy(No) = 0.0
 Entropy(Yes) = 0.0
 Gain(Student) = 0.971

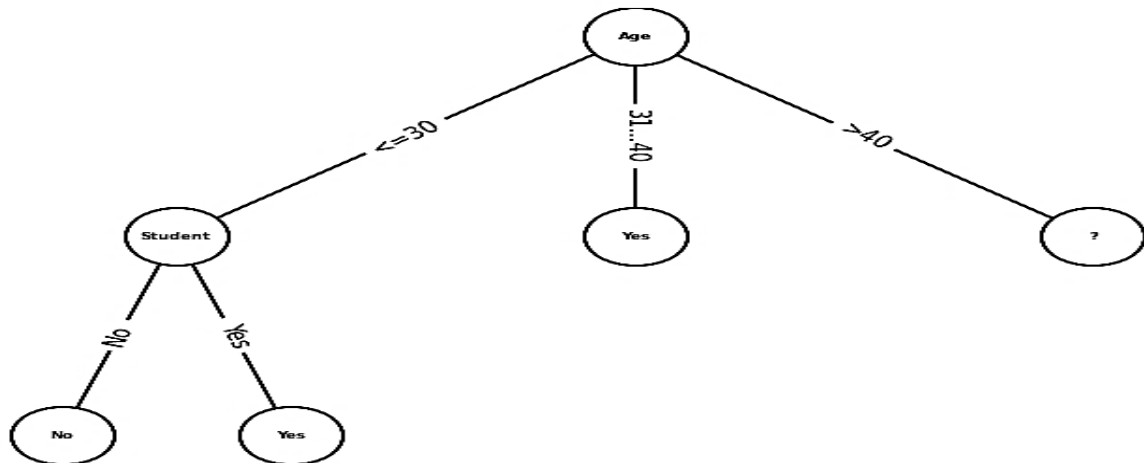
Attribute: Credit_Rating
 Entropy(Fair) = 0.9183
 Entropy(Excellent) = 1.0
 Gain(Credit_Rating) = 0.02

 Best Attribute Selected = Student

Tree Rules after iteration 2

IF Age = <=30 THEN:
 IF Student = No THEN Buys_Computer = No
 IF Student = Yes THEN Buys_Computer = Yes
 IF Age = 31...40 THEN Buys_Computer = Yes
 IF Age = >40 THEN ?

Decision Tree After Iteration 2



=====

ITERATION 3

=====

Current Node : Age = >40

CURRENT DATASET USED IN THIS ITERATION:

Age	Income	Student	Credit_Rating	Buys_Computer
>40	Medium	No	Fair	Yes
>40	Low	Yes	Fair	Yes
>40	Low	Yes	Excellent	No
>40	Medium	Yes	Fair	Yes
>40	Medium	No	Excellent	No

Entropy of Dataset = 0.971

Attribute: Income
 Entropy(Medium) = 0.9183
 Entropy(Low) = 1.0
 Gain(Income) = 0.02

Attribute: Student
 Entropy(No) = 1.0
 Entropy(Yes) = 0.9183
 Gain(Student) = 0.02

Attribute: Credit_Rating
 Entropy(Fair) = 0.0
 Entropy(Excellent) = 0.0
 Gain(Credit_Rating) = 0.971

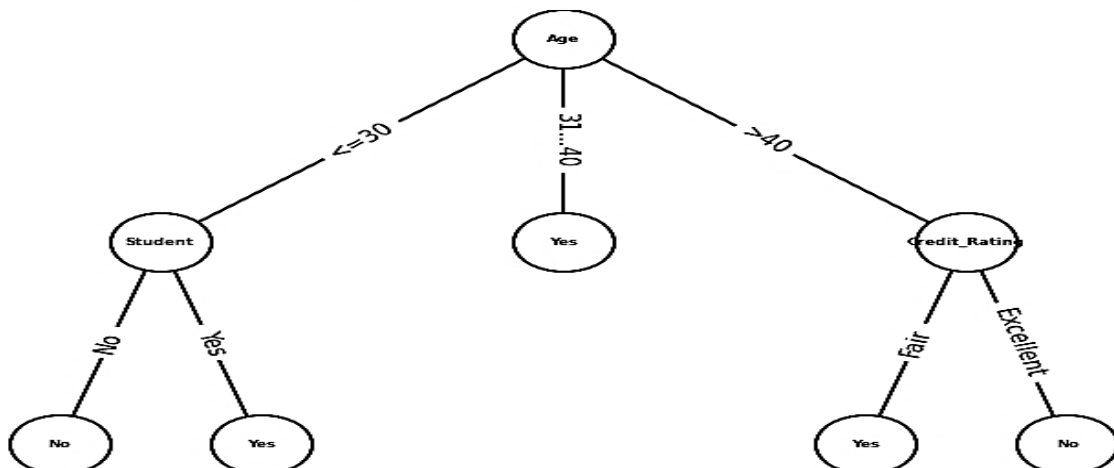
Best Attribute Selected = Credit_Rating

Tree Rules after iteration 3

```

IF Age = <=30 THEN:
  IF Student = No THEN Buys_Computer = No
  IF Student = Yes THEN Buys_Computer = Yes
IF Age = 31...40 THEN Buys_Computer = Yes
IF Age = >40 THEN:
  IF Credit_Rating = Fair THEN Buys_Computer = Yes
  IF Credit_Rating = Excellent THEN Buys_Computer = No
  
```

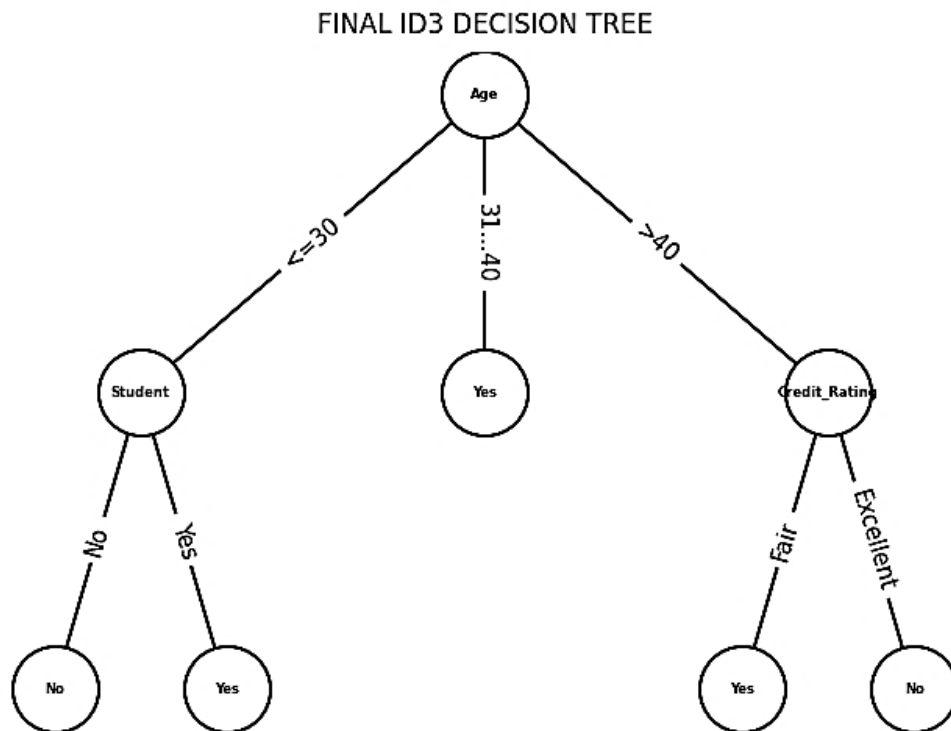
Decision Tree After Iteration 3



=====

FINAL DECISION TREE

=====



=====

ENTER VALUES TO CLASSIFY NEW SAMPLE

=====

Enter value for Age:<=30
 Enter value for Income:low
 Enter value for Student:yes
 Enter value for Credit_Rating:fair

=====

CLASSIFICATION RESULT

=====

New Sample: {'Age': '<=30', 'Income': 'low', 'Student': 'yes', 'Credit_Rating': 'fair'}
 Predicted Class: Yes